Loans and Credit: Testing the Institute's Beliefs

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Abstract

This report analyses Lending Club loan data to test three beliefs held by a Credit Institute, and to answer the Director's questions on default risk drivers, debt-to-income thresholds, and state-level differences.

1. Overview

This report analyses Lending Club loan data to test three beliefs held by the Credit Institute, and to answer the Director's broader questions on default risk.

A loan's outcome was classified as Fully Paid or Charged Off where resolved. Current, Late and Grace Period loans were excluded from default rate calculations, since their final outcome is not yet known. Across 383,202 resolved loans, the overall default rate is 22.5%, which serves as a baseline for comparison throughout this report.

2. Testing the Director's First Belief: Home Ownership and Employment Length

The Director's institute believes home owners and people employed 10+ years have lower default rates on short-term loans.

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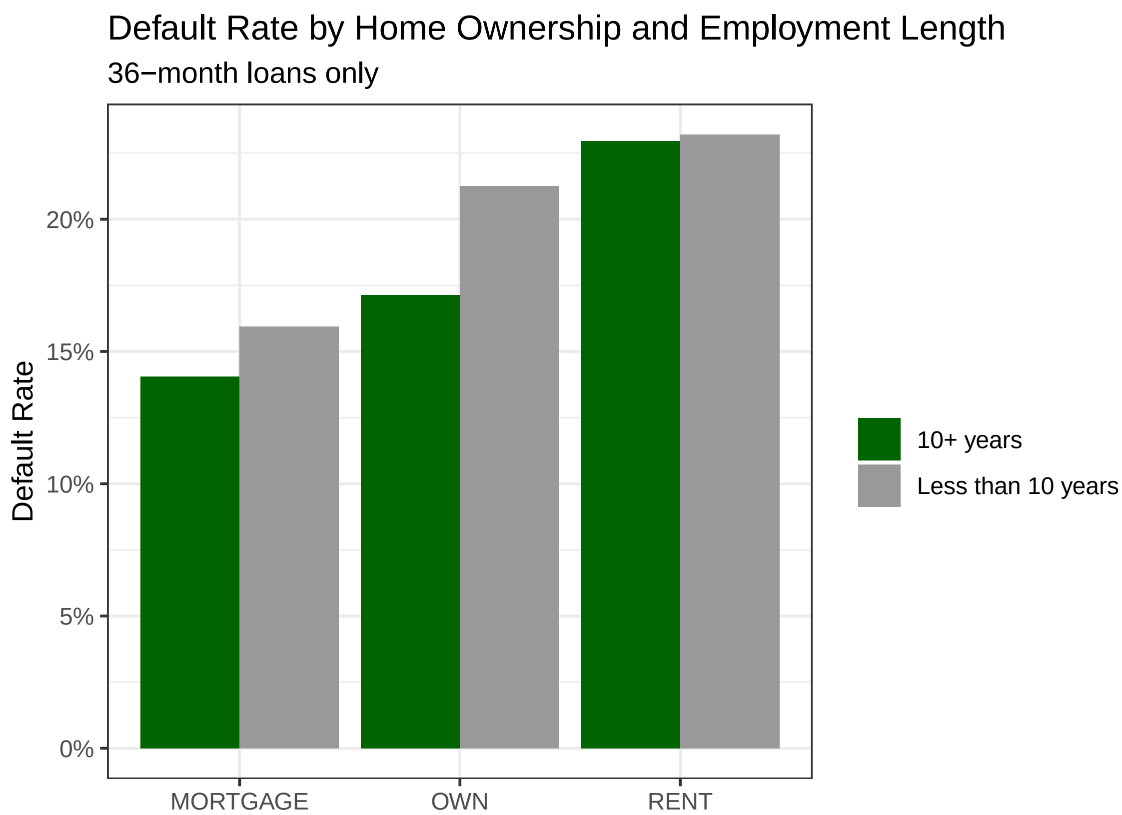
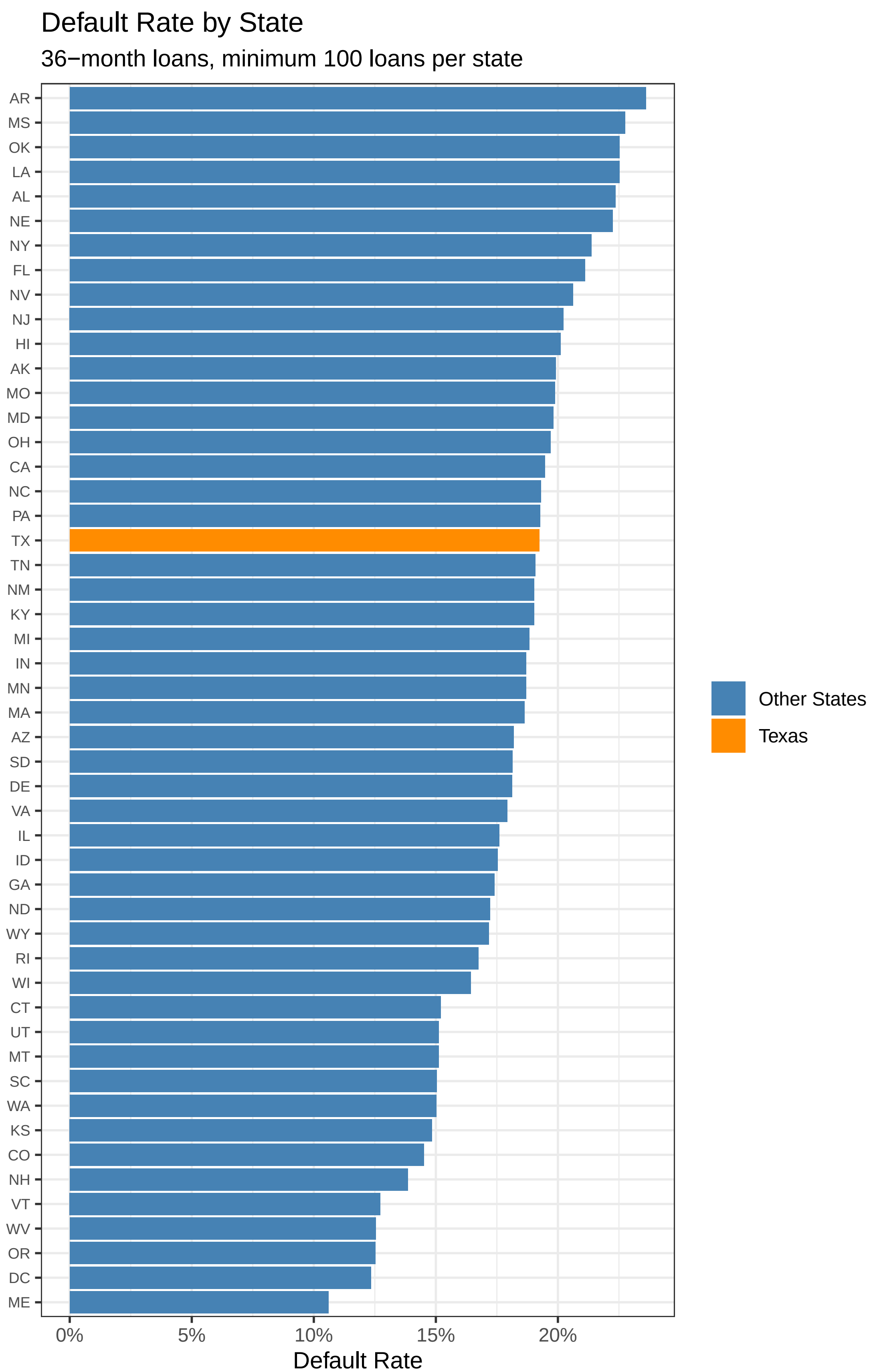


- Mortgage holders default least, at between 14.1% and 15.9%. Homeowners without a mortgage follow at 17.1% to 21.3%. Renters default most, at around 23%.
- Borrowers employed 10+ years default noticeably less than those with shorter employment histories, within both the mortgage and homeowner categories.
- Employment length makes little difference for renters. Both groups default at a similarly high rate. Housing status appears to be the stronger signal in this case.
- The Director's belief is well supported by the data, at least for short-term, 36-month loans.

3. Testing the Director's Second Belief: State-Level Default Differences

The Director's institute believes US States differ significantly in their culture of defaulting on short-term loans.


Default Rate by State
36−month loans, minimum 100 loans per state
AR
MS
OK
LA
AL
NE
NY
FL
NV
NJ
HI
AK
MO
MD
OH
CA
NC
PA
TX
TN
NM
KY
MI
IN
MN
MA
AZ
SD
DE
VA
IL
ID
GA
ND
WY
RI
WI
CT
UT
MT
SC
WA
KS
CO
NH
VT
WV
OR
DC
ME
0%
5%
10%
15%
20%
Default Rate
Other States
Texas

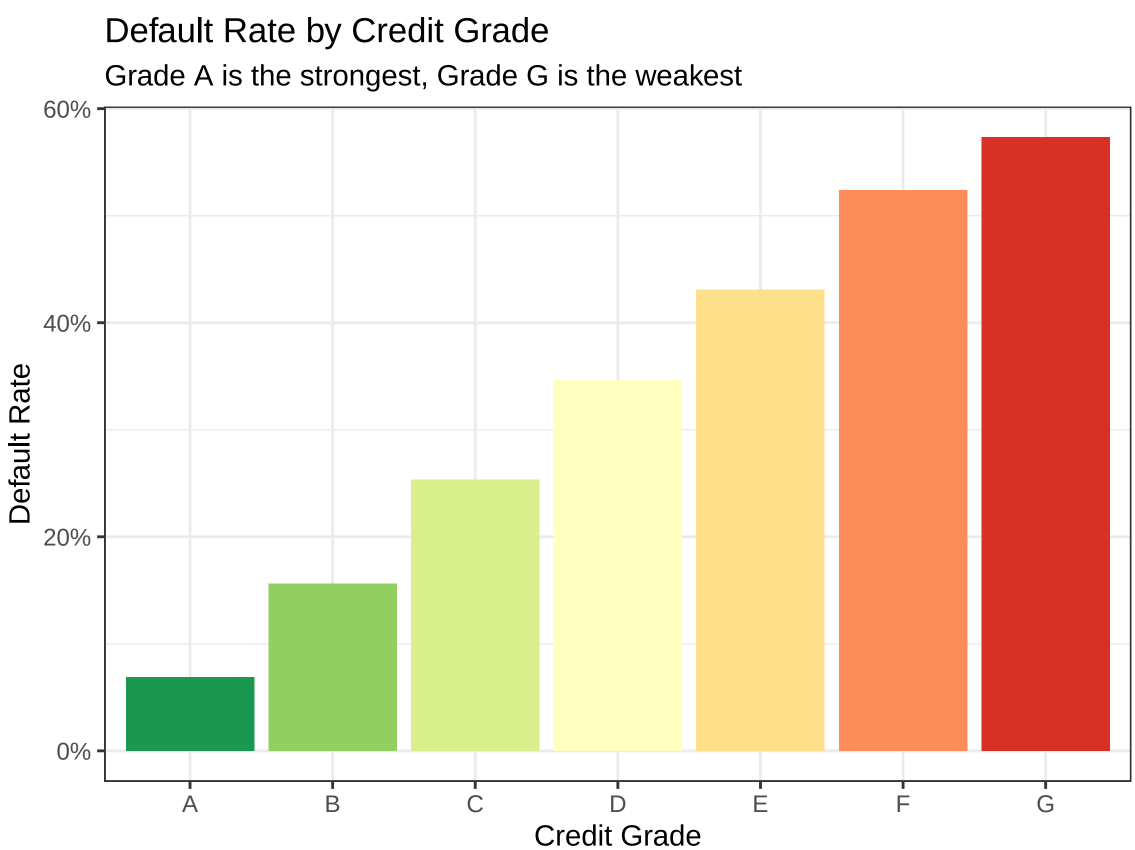
State default rates do genuinely differ. Arkansas sits highest at 23.6%, while Maine sits lowest at 10.6%. That gap is more than double, which supports the Director's belief.

A loose regional pattern emerges too. Several Southern states cluster near the top of the chart, including Arkansas, Mississippi, Oklahoma, Louisiana and Alabama. Several Northeastern and mountain states cluster near the bottom, including Maine, Vermont, West Virginia and Oregon. This pattern is worth noting, though it should not be read as proof of any particular cause.

Texas defaults at 19.2%, slightly below the national baseline of 22.5%. Texas sits roughly in the middle of the full state ranking, at position 19 of 50. Texas is not unusual compared to other states. It behaves much like a typical, average state in this dataset.

4. Testing the Director's Third Belief: Credit Grade and Interest Rate Determinants

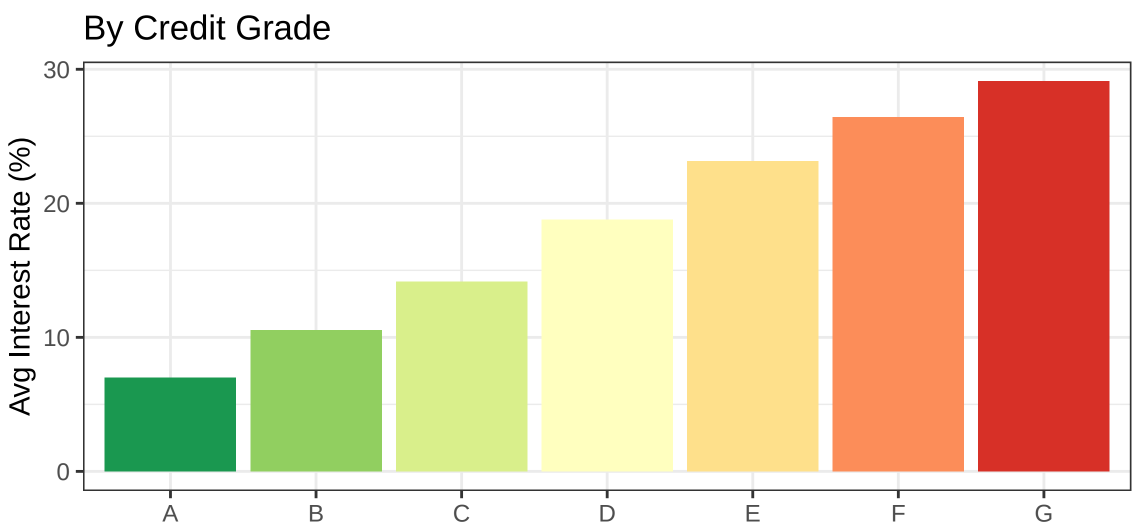
The Director's institute believes Credit Grades (A-F) are a great indicator of whether younger individuals will default, and that interest rates are determined by age, occupation and credit scores.

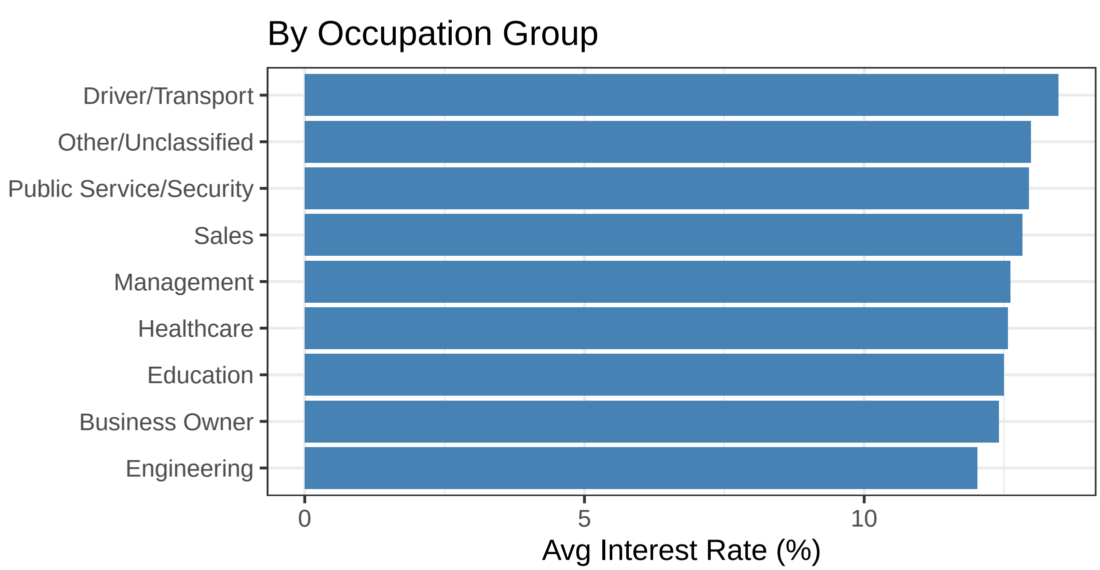


The Director's belief refers to grades A through F. The data also contains a seventh grade, G, which was included here since it follows the same pattern.

Credit grade predicts default very well. Default rates rise from 6.9% for Grade A to 57.3% for Grade G, in a near perfectly increasing staircase. This strongly supports the idea that grade is a great indicator of default risk.

One part of the belief could not be tested. The data contains no age column, so we cannot confirm whether grade is specifically a better or worse predictor for younger individuals.





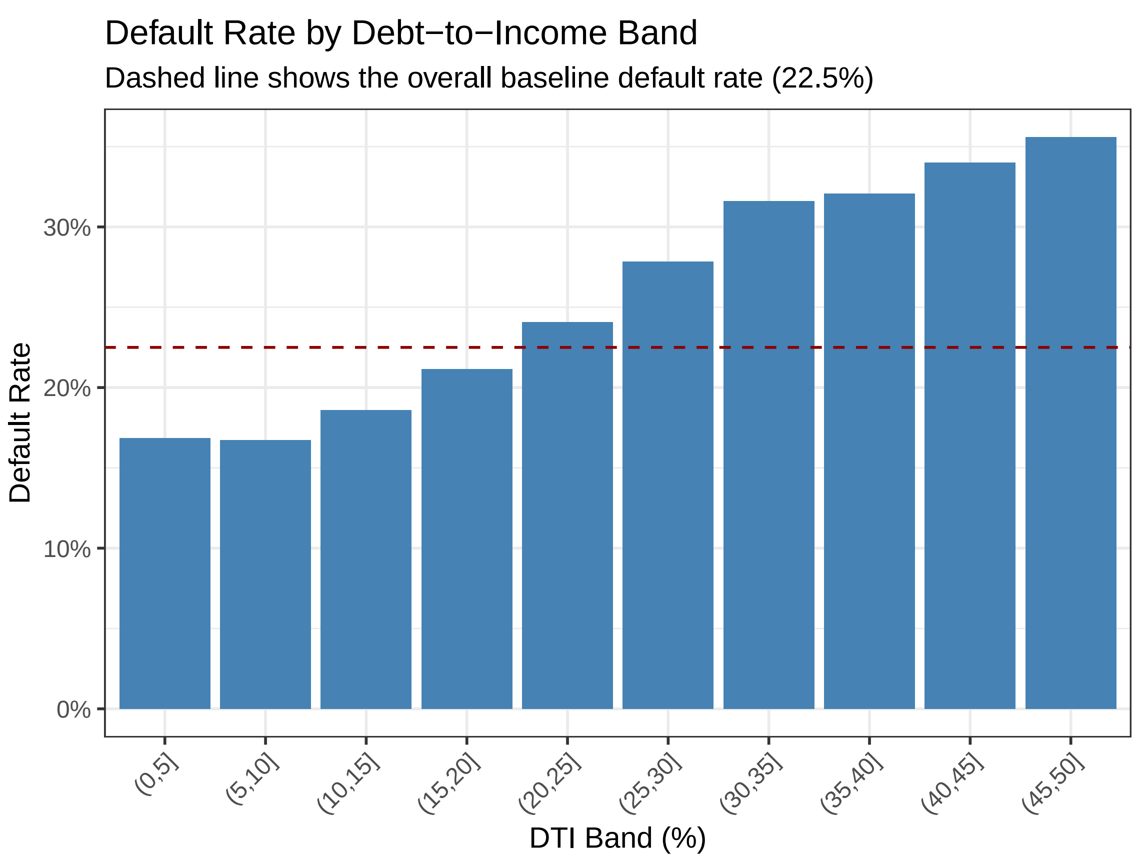
Credit grade strongly determines interest rate. Average rates rise from 7.0% for Grade A to 29.1% for Grade G, a spread of over 22 percentage points.

Occupation has only a weak relationship with interest rate by comparison. Average rates range from 12.0% for Engineering to 13.5% for Driver/Transport, a spread of just 1.5 percentage points.

Occupation data required substantial cleaning before this comparison was possible. Job titles were standardised for case and grouped into broad categories using keyword matching. Even after this effort, 47% of records could not be confidently classified into a specific occupation group and were kept as "Other/Unclassified". This limitation should be kept in mind when interpreting the occupation comparison.

Taken together, the data suggests credit grade is by far the dominant driver of interest rate. Occupation plays only a minor role. Age could not be tested at all, since no age variable exists in this dataset.

5. Debt-to-Income: Setting an Appropriate Cap



Default rate rises steadily and consistently with DTI. There is no single point where risk suddenly jumps. This means the right DTI cap depends on how much default risk the Institute is willing to accept, rather than one universally correct number.

Loans with DTI up to 20% default below the overall baseline rate of 22.5%. Loans with DTI above 20% cross into above-average risk territory, and the gap widens steadily from there.

For a conservative risk tolerance, capping DTI at 20% keeps default rate at or below 21.2%, modestly below the overall baseline.

For a moderate risk tolerance, capping DTI at 30% allows a default rate of around 27.9%, still well below the riskiest bands.

For an aggressive risk tolerance, allowing DTI up to 40% pushes default rate to around 32.1%, nearly 10 percentage points above baseline.

We would recommend a DTI cap in the region of 20% to 25% for most lending purposes, balancing reasonable access to credit against materially elevated default risk above this level.

The table below considers all loan terms, not only 36-month loans as in the earlier state ranking. This is why the default rates shown here differ slightly from those in the earlier section.

State	Avg DTI (%)	Default Rate	N Loans
AR	20.3	27.5%	2849
AL	20.6	26.3%	4688
TX	19.5	22.8%	32277
CA	17.0	22.5%	53705
VT	20.7	15.3%	767
ME	19.8	14.4%	1256

The relationship between DTI and default differs by state. Alabama's average DTI (20.6%) is actually higher than Maine's (19.8%), yet Alabama's default rate (26.3%) is nearly double Maine's (14.4%).

A single national DTI cap may not be equally appropriate everywhere. The same DTI level appears to carry meaningfully different real-world default risk depending on the state, echoing the broader state-level differences found earlier in this report.

6. Are the Institute's Heuristics Realistic?

Two of the three heuristics tested are well supported by the data. Home ownership and employment length genuinely relate to default risk. States genuinely differ in their default rates too.

The third heuristic is only partially realistic. Credit grade does predict default very well, which supports part of the claim. The data contains no age variable though, so the specific claim about younger individuals could not be tested at all. The claim that interest rates are clearly determined by occupation is also not well supported. Occupation showed only a small relationship with interest rate, especially compared to the much stronger relationship between credit grade and interest rate.

The Institute's heuristics are mostly grounded in real patterns overall. Some specific details, particularly around age and occupation, appear to be assumptions rather than findings confirmed by this data.

7. Summary and Recommendations

Key risk drivers for default, in order of strength:

- Credit grade is by far the strongest driver. Default rates rise from 6.9% for Grade A to 57.3% for Grade G.
- Debt-to-income ratio matters substantially too. Default rates rise steadily from around 17% at low DTI to over 35% at high DTI.
- Home ownership status is a meaningful driver. Mortgage holders default least, renters default most.
- Employment length adds further signal, particularly for mortgage holders and homeowners. Borrowers employed 10+ years default less often than those with shorter histories.
- State of residence matters too, with default rates ranging from 10.6% to 23.6% across states, for reasons not fully explained by DTI alone.
- Occupation, by contrast, shows only a weak relationship with risk-related outcomes once credit grade is accounted for.

Customers most likely to default:

The riskiest profile combines several of these factors at once: a low credit grade, a high DTI, renting rather than owning, less than 10 years in their current role, and residing in a higher-default state such as Arkansas, Mississippi or Oklahoma. Each factor compounds the others, so a borrower with several risk factors together is considerably riskier than any single factor would suggest alone.

What could reduce default risk in credibility assessment:

- Continue relying heavily on credit grade, since it remains the strongest available predictor in this dataset.
- Incorporate DTI more explicitly into assessment, given its steady and substantial relationship with default.
- Consider state of residence as a contributing factor, given the meaningful spread found across states that DTI alone does not explain.
- Treat occupation with caution as a standalone risk factor. The data does not support strong reliance on it once other factors are considered.
- Collect age data going forward. Its complete absence from this dataset prevented testing an important part of the Institute's existing risk heuristics.